

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System

Semester – Spring

Year : 2011
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

State the roles of DBA. What qualities should the DBA have to manage a very large database? 7

Define data model. Describe network and hierarchical data model. 8

What does QBE stands for? What are the different parts of SQL? 8
Enlist different data types used in SQL.

Consider following schema: 7

customer (cus_id, cus_name, cus_phno)

employee (cus_id, emp_id, emp_name, emp_add)

works (branch_id, salary, cus_id)

branch (branch_id, branch_name)

Write relational algebra notations for following queries for the given schema:

Select names of all employees.

Give a salary raise of 5% to all the employees.

Select names of all employees working for "Manang" branch.

List all branch names.

Delete any record from works table.

List names and phno of all customers.

Select names of all employees dealing with customer having id

3. a) What is integrity constraints? In how many categories the integrity constraints can be classified? Explain each. 8
- b) How redundancy is reduced in databases? Explain basic three steps giving an example. 7

OR

What do you mean by decomposition of relational schema? Explain lossless and lossless decomposition with example. 7

4. a) What is cryptography? Explain the type of cryptography system. 7
- b) Define equivalence of expression? State steps used for query optimization by algebraic manipulation. 8
5. a) Explain sequential file organization. What are hash functions explain giving example. 8
- b) Explain indexing and B-tree file structure. 7

OR

What is data dictionary storage? Mention the information's which are kept in it by defining the categories. 7

6. a) What is crash recovery? Explain about shadow paging techniques. 7
- b) What is a transaction? Explain concurrency control protocols. 8
7. Write short notes on **any two**: 2×5
 - a) Multiple granularity
 - b) Distributed database model
 - c) Multi-valued and join dependencies

POKHARA UNIVERSITY

Level: Bachelor

Semester – Fall

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Attempt all the questions.

1. a) Define information. Explain the statement "DBA has central control over the database system." 7
- b) Construct an ER diagram for keeping records for exam section of a college. 4
- c) Explain the distinction among the terms primary key, candidate key, super key and foreign key with an example. 4
2. a) Consider the following database : 8
Student(sid, name, age)
Has(sid, cid)
College(cid, cname)
Write relational algebra expression to perform the following:
 - i. Find average age of student
 - ii. Display name of student who studies in "QWERT" college.
 - iii. Insert a new student.
 - iv. Delete record of "ASDFG" college from college relation.
 - v. Display name of students whose name begin from 'S'.
- b) Consider the relational database 7
Employee (empname, street, city)
Works (empname, cmpname, salary) Company (cmpname, city)
Manages (empname, cmpname)
Write SQL statement to:
 - i. Modify the database so that Amrit now lives in Naxal
 - ii. Delete all tuples in the works relation for employee of xyz corporation
 - iii. Increase salary of all employees of ABC company by 10%

- iv. Display all company name located at city Pokhara or Kathmandu from the company tables.
 - v. Display all empname who have salary greater than 5000 from works table
3. a) State and explain about functional dependency. Considering a suitable example.

OR

What is normalization? Explain about BCNF and 4th normal forms in detail by taking example.

- b) Define third normal form. Convert the following 2NF relation into 3NF (consider **Name** as primary key)

<u>Name</u>	Address	Phone	Salary	Post
Gill	KTM	456789	20000	Engineer
Van	BKT	654321	20000	Engineer
Robert	KTM	456789	20000	Engineer
Brown	BKT	654321	10000	Overseer
Albert	KTM	454545	10000	Officer

4. a) Is it necessary to manage security at OS level if security in database level is already done? Explain private key cryptosystem.
- b) What is query processing? Explain in detail each step involved in it
5. a) State and explain B- tree index. How indexes improves the performance of file storage?
- b) Describe about failure classification? Write down Differences between Deferred Database Modification and Immediate Database Modification.
6. a) What do you mean by a schedule? When schedule is called serializable? What are conflict serialization schedules?
- b) What is OODBMS. Discuss Distributed database along with necessary diagram.
7. Write Short notes on **any two**:
- a) Buffer management
 - b) Multivalued Dependency
 - c) Time-stamping based protocol

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3. a)

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Attempt all the questions.

“Database approach is more appropriate than file-processing system for application development”. Give reasons in the support of this statement. 8

Define data model. Explain the different data model. 7

Consider the following relations for an order-processing database application in a company. 8

CUSTOMER (Cust#, Cname, City)

ORDER(Order#, Odate, Cust#, ord_Amt) 4.

ORDER_ITEM(Order#, Item#, Qty)

ITEM(Item#, Unit_Price)

SHIPMENT(Order#, Wavehouse#, Ship_date) 5.

WAREHOUSE(Warehouse#, City)

Answer the following queries in relational Algebra

i. List the order# and ship date for all orders shipped from wavehouse number “W2” 6.

ii. List the warehouse information for which the customer named ‘Jose Copez’ was supplied his orders 7.

iii. List the orders that werenot shipped within 30 days of ordering

iv. List the order # for orders that were shipped form all werehouses that the company has in network.

5) What is Integrity constraint? How does data integrity Constraint differ with Data Security. 7

3. a) Consider the following relations:

Employee (empID, FirstName, LastName, address, DOB, sex, position, deptNo)

Department (deptNo, deptName, mgr, empID)

Project (projNo, projName, deptNo)

Work on (empID, projNo, hours worked)

Write the SQL statements for the following:

- i. List the name and addresses of all employees who work for the IT department.
- ii. List the total hours worked by each employee, arranged in order of department number and within department, alphabetically by employee surname.
- iii. List the total number of employees in each department for those departments with more than 10 employees.
- iv. List the project number, project name and the number of employees who work on that project.

- b) What is integrity constraints? Explain with example about BCNF and 5th normal form.

4. a) Diagrammatically illustrate and discuss the steps involved in processing a query.

- b) Explain the major issues related to Database security.

5. a) What is data dictionary storage? Explain Heap file organization.

- b) What is file organization? Explain heap file organization with its advantages and disadvantages over index file organization.

6. a) What do you understand by concurrency control? Discuss two phase locking protocol. How does it guarantee serialization.

- b) What is log-based re-covery? How is it different from shadow paging.

7. Write short notes on: (Any two)

- a) B-tree index file.

- b) Distributed DBMS.

- c) Natural join.

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Attempt all the questions.

1. a) The DBA controls the security aspects of the DBMS implementation. Discuss the software and hardware controls available to the DBA, and how this could fit into a commercial environment. 7
b) What is super key and candidate key? Explain in brief about the structure of RDBMS. 8
2. a) What do you mean by relational algebraic operators? Explain all the basic operators with examples. 8
b) Explain DDL and DML operations with suitable example. 7
3. a) State and explain in brief about multi-valued and joined dependencies. 8
OR
What is normalization? State and explain in brief about 4th and 5th normal form with suitable example.
b) State ACID rules for concurrency control. Explain lock-based protocols. 7
4. a) What is integrity violation? Discuss the security levels that can be applied in DBMS. 8
b) Define query optimization? Explain in brief about equivalence of expression. 7
5. a) Explain the structure of Index sequential file with the help of a diagram. 7
b) What is stable storage? Explain in brief about shadow paging. 8
6. a) What is transaction? Describe the dead lock handling mechanism. 8
b) Explain entity integrity and referential integrity. Also, give an example of each. 7

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7. Write short notes on: (Any two)

- a) Data Dictionary.
- b) Distributed Model.
- c) Denormalization.

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Attempt all the questions.

1) Why an organization needs DBMS? Explain three levels of data abstractions. 7

2) Construct an ER diagram for a departmental store that keeps the information about customer, supplier and items. Associate with a log of various selling and purchasing information. The ER diagram should explain Entity name, some appropriate attributes of the entity, relationship among the entity, association and directivity among the entities. 8

3) Consider the following schemas of a relational database: 8

Employee (emp_name, street, city)

WorksIn (emp_name, company_name, salary)

Company (company_name, city)

Manages (emp_name, manager_name)

Write relational algebra for

i. Find the names and cities of residence of all employers who work for "Nabil Bank".

ii. Find the name, street address and cities of residence of all employers who work for "Nabil Bank" and earn more than 2,40,000/- per annum".

iii. Modify the database so that "Kiran" now lives in "Kathmandu".

iv. Give all employees of "Nabil Bank" a 10 percent salary rise. 7

4) What are Query Languages, explain about its importance in DBMS. Differentiate between DDL and DML. 7

5) What do you mean by Functional Dependency, Multivalued Dependency and Transitive Dependency in Normalization process of Database? Why normalization is needed? Assume the un-normalized relation as given below and find the final normalized logical ER 10

3. a) Consider the following relations:
- Employee (empID, FirstName, LastName, address, DOB, sex, position, deptNo)
 - Department (deptNo, deptName, mgr, empID)
 - Project (projNo, projName, deptNo)
 - Work on (empID, projNo, hours worked)
- Write the SQL statements for the following:
- i. List the name and addresses of all employees who work for the IT department.
 - ii. List the total hours worked by each employee, arranged in order of department number and within department, alphabetically by employee surname.
 - iii. List the total number of employees in each department for those departments with more than 10 employees.
 - iv. List the project number, project name and the number of employees who work on that project.
- b) What is integrity constraints? Explain with example about BCNF and 5th normal form.
4. a) Diagrammatically illustrate and discuss the steps involved in processing a query.
- b) Explain the major issues related to Database security.
5. a) What is data dictionary storage? Explain Heap file organization.
- b) What is file organization? Explain heap file organization with its advantages and disadvantages over index file organization.
6. a) What do you understand by concurrency control? Discuss two phase locking protocol. How does it guarantee serialization.
- b) What is log-based re-recovery? How is it different from shadow paging.
7. Write short notes on: (Any two)
- a) B-tree index file.
 - b) Distributed DBMS.
 - c) Natural join.

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Attempt all the questions.

List out the major objectives of DBMS. Differentiate between Data abstraction & Data independence. 7

Draw an E-R diagram for the database of a hospital with a set of patients and a set of medical doctors. With each patient a log of the various tests conducted is also associated. Make your own assumptions if necessary. 8

What is relational algebra? Compare and contrast relational algebra with the relational calculus. 7

Consider following relations: 8

employee (emp_name, street, city)
 works (emp_name, company, salary)
 company (comp_name, city)
 manages (emp_name, manager_name)

Write SQL statements for:

- i. Find employee names that lives in the city same as the company city.
- ii. List all employee details who earn more then 25000.
- iii. Update address of an employee 'Sriyash' to 'Pokhara'.
- iv. Create a view for which employee earns Rs. 20,000 or more.
- v. Delete all employees from the table employee.

Explain BCNF and 3NF with suitable example. 7

Differentiate between authentication & authorization. How encryption & decryption occurs in Private key & Public key cryptography? 8

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4. a) How the query optimization process is carried out? Explain about cost estimation of query. 8
- b) When is it preferable to use a dense index rather than a sparse index? Explain with a suitable example. 7
5. a) Discuss in detail about the shadow paging technique of crash recovery with its drawbacks. 7
- b) Define dead lock and serializability. Illustrate dead lock and conflict serializability with suitable example. 8
6. a) Under which situations will be beneficial to have replication or fragmentation of data? Explain with suitable example. 8
- b) Define database integrity. Classify the integrity constraints of database. 7
7. Write short notes on: **(Any two)** 2×5
- a) Assertions and Trigger
- b) 2PL
- c) Stored procedure.
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Attempt all the questions.

1. a) What are the major responsibilities of Database Management System? 7
For each responsibility, explain the problems that would arise if the responsibility were not discharged 4.
b) Compare Hierarchical Model, Network Model and Relational Model base on operation performed. 8
2. a) Consider the relational database of Figure below, Where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries: 8
Employee (person-name, street, city) 5.
Works (person-name, bank-name, salary)
Bank (bank-name, city)
Manages (person - name, manager- name)
i. Find the total salary sum of all the banks.
ii. Modify the database so that Ram now lives in Kathmandu.
iii. Find the name, street address, and cities of residence of all employees who work for Nepal World Bank Corporation and earn more than \$10,000 per annum. 6.
iv. Delete all tuples in works relation for employee of Small Bank Corporation 7.
b) State and explain B+ tree index. Show an example of Insertion on B+ trees. 7
3. a) Consider the employee database of figure given below, where primary keys are underlined. Give an expression in SQL for each of following queries. 10

Employee (employee-name, street, city)

Works (employee-name, company-name, salary
company (company-name, city)

manages (employee-name, manager-name)

- i. Modify the database so that Ram now lives in Kathmandu.
 - ii. Give all employees of First Bank Corporation a 10 percent raise.
 - iii. Give all managers of First Bank Corporation a 10 percent raise.
 - iv. Delete all tuples in the works relation for employees of small Bank Corporation.
 - v. Find all employees who earn more than the average salary of all employees of their company.
- b) Explain the need of stored procedures & its application.
4. a) What is referential integrity? Explain the functional dependency and multivalued dependency with suitable example.
- b) What is Normalization and why it is done? Give an example of a relation schema R and a set of dependencies such that R is not in 3NF and normalize it into 3NF.
5. a) What are the needs of cryptography? How security can be granted using view? Explain?
- b) Show how to drive the following equivalences by a sequence of transformations using the equivalence rules.
- i. $\sigma_{\theta_1 \wedge \theta_2 \wedge \theta_3}(E) = \sigma_{\theta_1}(\sigma_{\theta_2}(\sigma_{\theta_3}(E)))$
 - ii. $\sigma_{\theta_1 \wedge \theta_2 \bowtie \theta_3}(E_1 \bowtie E_2) = \sigma_{\bowtie}(E_1 \bowtie \sigma_{\theta_2}(E_2))$, where θ_2 involves only attributes from E_2
6. a) What do you mean by shadow paging? Explain Deferred Database modification with an illustration.
- b) Explain the concept of locking for concurrency control.
7. Write short notes on: **(Any two)**
- a) ORM
 - b) 2PL
 - c) Hash Index

POKHARA UNIVERSITY

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Attempt all the questions.

Compare & Contrast between file system & Database system. 7

Difference between Schema & Instances.

Define data model. Explain the different types of data model. 8

Consider the relational database of Figure below, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries: 8

employee (person-name, street, city)

works (person-name, bank-name, salary)

bank (bank-name, city)

manages (person-name, manager-name)

i. Find the names of all employees who work for Nepal Rastra Bank and Salary greater than \$10,000.

ii. Find the names and cities of residence of all employees who work for Nepal Rastra Bank.

iii. Find the names, street address, and cities of residence of all employees who work for Nepal Rastra Bank Corporation and earn more than \$10,000 per annum.

iv. Delete all tuples in works relation for employee of Nepal Rastra Bank.

b) Write SQL statements for the following queries in reference to relation **Emp_time** provided. 7

Eid*	Name	Start_time	End_time
E101	Hari	10:15	18:00
E102	Malati	8:00	15:30

E103	Kalyan	9:30	17:00
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- i. Create the table **Eid*** as primary key and insert the values provided.
 - ii. Display the name of the employee whose name start from letter 'M' and who work for more than seven hours.
 - iii. Delete the entire contents of the table so that new data can be inserted.
3. a) What do you mean by Normalization? Explain the BCNF and 5th normal form with examples 8
- b) Differentiate between authentication & authorization. How encryption & decryption occurs in Private key& Public key cryptography? 7
4. a) What are the basic steps in Query Optimization? Explain with suitable diagram. 8
- b) List out the major advantages of B+ tree index. Explain the concept of Hashing. 7
5. a) Compare the shadow paging recovery scheme with the log based recovery scheme. 8
- b) What do you mean by a schedule? When schedule is called serializable? What are conflict serialization schedules? 7
6. a) Explain the roles of Assertions and Triggers in SQL. When Triggers are not appropriate to use? Give an example 8
- b) Explain the distinction among the terms primary key, candidate key, super key and foreign key with an example. 7
7. Write short notes on: (**Any two**) 2×5
 - a) Applications of Database
 - b) Functional Dependency
 - c) Distributed Database

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Attempt all the questions.

1. a) Explain the importance of DBMS in current world. How is it accessed using various DDL, DML and DCL languages?
- b) Suppose you are given the following requirements for a simple database for the National Cricket League (NCL):
 - the NCL has many teams
 - each team has a name, a city, a coach, a captain, and a set of players
 - each player belongs to only one team
 - each player has a name, a type (such as batsman or bowler), a skill level, and a set of records
 - a team captain is also a player
 - each player is sponsored by at least one brand
 - a brand has its name, established date, property, multiple contact_no.

Construct a clean and concise ER diagram for the NCL database. List your assumptions and clearly indicate the cardinality mappings as well as any role indicators in your ER diagram.

2. a) Consider the relational database of Figure below, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries:
employee (person-name, street, city)
works (person-name, bank-name, salary)
bank (bank-name, city)
manages (person-name, manager-name)

- i. Find the names of all employees in this database who live in the same city as the company for which they work.
- ii. Give all employees of First Bank Corporation a 10 percent salary raise.
- iii. Modify the database so that Harish now lives in Biratnagar.
- iv. Delete all tuples in works relation for employee of First Bank Corporation.

b) Construct a B+-tree for the following set of key values:

(2, 3, 5, 7, 11, 17, 19, 23, 29, 31)

Assume that the tree is initially empty and values are added in ascending order. Construct B+-trees for the case where the number of pointers that will fit in one node is **Four**. Also show the form of the tree after deletion of 23.

3. a) Suppose we are given the following table definitions with certain records in each table. (Underline attribute represent primary key attributes.).

EMPLOYEE (EID, NAME, POST, AGE)

POST (POST-TITLE, SALARY)

PROJECT (PID, PNAME, DURATION, BUDGET)

WORK-IN (PID, EID, JOIN-DATE)

Write the SQL statement for

- i. List the name of employees whose age is greater than the average age of all employees.
- ii. Display all employee numbers of those employee who are not working in any project
- iii. List name of employee and their salary who are working in the project "DBMS".
- iv. Update the database so that "Rishab" now lives in "Butwal".

b) What is joining in DBMS? Explain different types of join with example.

4. a) What do you mean integrity constraints? Explain assertion and triggers in SQL with their syntax.

b) Define functional dependency. Consider a table which is in 2NF but not in 3NF. Break the table so that it is now in 3NF with a table example.

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Attempt all the questions.

Define RDBMS. Explain the differences between file oriented system and a database oriented system. 7

Construct E-R model for a car insurance company whose costumer own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Also Design a relational database corresponding to the E-R diagram. 8

Consider a student registration database comprising of the below given schema. 6

Student(CRN, Name, Gender, Address, Telephone)

Course(CourseID, CourseName, Hour, TeacherID)

Teacher(TeacherID, TeacherName, Office) .

Registration(CRN, CourseID, Date)

Write relational algebraic expression for the following tasks:

- Count the number of student registered subject in year 2015 gender wise.
- Show student details taught by teacher Rohit Shrestha.
- Delete student information taught by teacher N. Mathema.

Consider a relational Schema:

Teacher(TeacherID, TeacherName, Office)

Write SQL statement for the following task:

- To create a table from a table.
- To eliminate duplicate rows.
- To add a new column 'Gender' in the table.
- To sort data in a table.
- To delete rows.

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Attempt all the questions.

1. a) Explain the need of DBMS over file system. Explain the various level of data abstraction with examples. 7
- b) Construct an ER diagram for a banking enterprise that keeps the information about employee, customer, loan, account and payment. 8
2. a) How does a view differ with relation? Define the role of view in security. 7
- b) Consider the following schema of a relational database. 8
Branch (branch-name, branch-city, assets)
Account (account-number, branch-name, balance)
Customer (customer-id, customer-name, customer-street, customer-city)
Depositor (customer-id, account-name)
Loan (loan-number, branch-name, amount)
Borrow (customer-id, loan-number)
Write the relational algebra for the following queries:
 - i. Find all customer either account or loan
 - ii. List the name and city of customer who have their account at the branch location 'Butwal'.
 - iii. Delete all account in the branch "B1"
 - iv. Increase balance by 5% to all branches
3. a) Define stored procedure. List the advantages and disadvantages of stored procedure. Explain how stored procedure are created with example. 7
- b) Consider a simple relational database of Hospital Management System. (Underlined attributes represent Primary key attributes) 8
Doctors (DoctorID, DoctorName, Department, Address, Salary)

Patients (PatientID, Patient Name, Address, Age, Gender)

Hospitals (PatientID, Doctor ID, HospitalName, Location)

Write Down the SQL statement for the following.

- i. Display ID of Patient admitted to hospital at Pokhara and whose name ends with 'a'.
 - ii. Delete the record of Doctors whose salary is greater than average salary of doctors.
 - iii. Increase the salary of doctors by 18.5% who works in OPD department.
 - iv. Find the average salary of Doctors for each address who have average salary more than 55K.
4. a) What do you mean integrity constraints? Explain assertion and triggers in SQL with their syntax. 7
- b) Define functional dependency. Explain BCNF and 3NF with suitable examples. 8
5. a) Construct a B+ tree for the following set of key values: 8
{2, 3, 5, 11, 17, 19, 23, 29, 31}
Assume that the tree is initially empty and values are added in ascending order.
- b) What do you mean by query processing? Explain the query optimization process. 7
6. a) What is log? Discuss the salient features of deferred database modification and immediate database modification strategies. 7
- b) Differentiate between exclusive lock and shared lock. Discuss the conflict and view serializability with suitable example. 8
7. Write short notes on: (**Any two**) 2×5
 - a) Remote backup system
 - b) Distributed database
 - c) Cryptography

vi. Count number of rows based on Office.

3. a) State the need for Normalization of a database and explain the 1NF, 2NF and 3NF with suitable example.
b) What is functional dependency? Explain its types in detail.
4. a) Explain sequential file organization. What are hash functions, explain giving example.
b) Discuss about the Access control mechanisms and cryptography methods to secure the database.
5. a) Write a detail description about Query Processing and Optimization. Explain the cost estimation of Query Optimization.
b) Difference between object oriented model distributed database.
6. a) Compare the Shadow paging recovery scheme with the log based recovery schema.
b) Explain the conflict and view serializability with suitable example. Discuss the testing of serializability also.
7. Write short notes on: (**Any two**)
a) DDL and DML SQL statement
b) ACID Properties
c) Stored procedure.

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Attempt all the questions.

Describe about Schemas and Instances Write briefly about DDL and DML. 7

Draw an ER diagram for the following scenario.

A university contains many faculties. The faculties in turn are divided into several colleges. Each college offers numerous programs and each program contains many courses. Teachers can teach many different courses and even the same course numerous times. Courses can also be taught by many teachers. A student is enrolled in only one program but a program can contain many students. Students can be enrolled in many courses at the same time and the courses have many students enrolled. 8

1) Consider the following schema: 8

employee (person_name, street, city)

works (person_name, company_name, salary)

company (company_name, city)

manages (person_name, manager_name)

Give an expression in relational algebra to express each of the following queries:

- Find the names of all employees who earn more than their managers
- Find the names of all employees who live in the same city and on the same street as their managers
- Find the names of all employees within the database that do not work for "NBL company"
- Find the names of all employees in the database who earn

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2017

Programme: BE

Full Marks: 100

Course: Database Management System

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define database management system (DBMS). Mention the advantages of DBMS. Explain data independence with its importance. 7
b) What do you mean by data model? What are the basic data modelling components? Briefly explain different types of data models. 8
2. a) Define relation schema and views. Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course: 8
STUDENT(SSN, Name, Major, Bdate)
COURSE(Course#, Cname, Dept)
ENROLL(SSN, Course#, Quarter, Grade)
BOOK_ADOPTION(Course#, Quarter, Book_ISBN)
TEXT(Book_ISBN, Book_Title, Publisher, Author)
Draw a relational schema diagram specifying the foreign keys for this schema.
b) Explain several parts of Structured Query Language (SQL). What are the basic domain types? Describe them. 7
a) Describe the basic structure of SQL queries. Considering at least two relations, write SQL for illustrating different types of set operations. 7
b) Design relational database for the Dept. of Computer Engineering (DoCE) at Pokhara University. Your database should have at least three (3) relations. Describe referential integrity constraint based on the above database of DoCE. 8
4. a) Define normalization in database. Mention its significances. With example, explain requirements to satisfy 1NF, 2NF, and 3NF. 8
b) Briefly explain encryption techniques to secure application data. 7

5. a) With diagram, briefly explain the basic steps of query processing. 7
b) Define indexing in database. With example, describe the structure of a B⁺-tree. 8
6. a) Explain the architecture of remote backup system. Discuss several issues that must be addressed while designing it. 8
b) Define transaction and explain its ACID properties. Describe the two-phase locking protocol for concurrency control. 7
7. Write short notes on: (**Any two**) 2×5
a) Data Dictionary
b) QBE
c) Functional Dependencies

more than the top earner at "NBL Company" in the database.

- b) Write the SQL statements for the following queries by reference of **Liquors_Info** relation:

Serial No	Liquors	Start year	Bottles	Ready year
1	Gorkha	1997	10	1998
2	Divine Wine	1998	5	2000
3	Old Durbar	1997	12	2001
4	Khukuri Rum	1991	10	1992
5	Xing	1994	5	1995

- i. Create the Liquors_Info relation.
 - ii. Insert the records in Liquors_Info as above.
 - iii. List all the records which were ready by 2000.
 - iv. Remove all records from data base that required more than 2 years to get ready.
3.
 - a) How does "GROUP BY" clause work? What is the difference between WHERE and HAVING clause? Explain each with examples
 - b) What is a database anomaly? Explain different types of database anomalies with suitable examples.
4.
 - a) What do you mean by normalization process? Why is it necessary in RDMBS? Justify.
 - b) Differentiate between authorization and authentication with brief examples.
5.
 - a) Why ACL technique is considered safe- way for database security? How is any user allowed or prevented from accessing a certain resource? Justify technically.
 - b) What is Query optimization? How can it be achieved?
6.
 - a) Explain how records of a file are placed and organized into a secondary storage.
 - b) What is Remote backup system? How does it help any organization? Clarify.
7. Write short notes on: (Any two)
 - a) ACID Properties of transaction
 - b) Concurrency control
 - c) Distributed Databases

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Explain the difference between physical and logical data independence. List the major steps that you would take in setting up a database for a particular enterprise. 7
- b) Suppose you are given the following requirements for a simple database for the Employee Management System: 8
- An employee may work in up to two departments or may not be assigned to any department. 7
 - Each department must have one and may have up to three phone numbers. 7
 - Each department can have anywhere between 1 and 30 employees. 8
 - Each phone is used by one, and only one, department. 7
 - Each phone is assigned to at least one, and may be assigned to up to 30 employees.
 - Each employee is assigned at least one, but no more than 5 phones.

Construct a clean and concise ER diagram for the database. Clearly indicate the cardinality mappings. 7

2. a) Consider the following relational Schema:

Department (**DepartmentID**, DepartmentName)
Designation (**DesignationID**, DesignationName, Salary)
Employee (**EmpID**, mpName, Gender, DesignationID, DepartmentID)
Allowance (**AllowanceID**, AllowanceName)
Allowance Details (DetailID, EmpID, AllowanceID, Amount)

Write the relational algebraic expression for the following task:

- i. Find the number of employees department-wise.
 - ii. List the employee details whose total salary is above Rs. 50000.
 - iii. List the employee those who are getting house allowance.
- b) Consider the following three relations. 8

Doctor(Name, age, address)

Works(Name, Depart_no, salary)

Department(Depart no, depname, floor, room)

Write down the SQL statement for the following.

- i. Display the name of doctor who do not work in any department.
 - ii. Modify the database so that Dr. Hari lives in Pokhara.
 - iii. Delete all record of Doctor working in OPD department.
 - iv. Display the name of Doctors who work in at least two departments.
3. a) Differentiate between SQL and MYSQL. Why access to database from a general purpose programming language is required? Explain. 7
- b) Define 1NF, 2NF and 3NF. What is the motivation behind normalizing the database? 8
4. a) What are the roles of Assertions and Triggers in SQL? Consider following bank database: 7
- Branch-schema = (branch-name, branch-city, assets)
Loan-schema = (loan-number, branch-name, amount)
Write an assertion for the bank database to ensure that the Assets value for the Koteswor branch is equal to the sum of all the amounts lent by the Koteswor branch.
- b) Why security is needed in database? How security can be granted using view explain. 8
5. a) Construct a B+-tree for the following set of key values: 7
- (1, 3, 6, 7, 11, 17, 19, 23, 30, 32). Assume that the tree is initially empty and values are added in ascending order.
- Construct B+-trees for the case where the number of pointers that will fit in one node is Four. Also show the form of the tree after

insertion of 9.

- b) Design an E-R diagram for keeping track of the exploits of your favourite sports team. You should store the matches played, the scores in each match, the players in each match and individual player statistics for each match. Summary statistics should be modeled as derived attributes. 8
6. a) What is conflict serializable? Explain the concept of locking for concurrency control. 7
- b) What are the various types of failures that can occur in database? Discuss the log based recovery mechanism. 8
7. Write short notes on: (**Any two**) 2×5
- a) ACID property
- b) ORM
- c) Network Data Model Vs Hierarchical Data Model

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System

Semester: Spring

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Considering an example, differentiate between data and information. Explain, how DBMS overcome the limitations of traditional file processing system.
- b) Construct an *ER diagram* for a Metropolitan Bus Park. There are many gates for entering bus park. Different gates are assigned to different routes. A route uses different buses. Bus consists of different seats which are assigned to different passengers. Frequent travelers are also in passenger. Associate a log of reservation date while reserving seats. The passenger name must have two attributes first_name & last_name. Each of the entities must have primary key attribute as far as possible. The cardinality mappings should be explained properly.

- a) Consider the relational database model:

Users (uid, cname, city)

Items (itemid, itemname, city, quantity, price)

Manager (mid, aname, city)

Query (queryno, uid, mid, itemid, query_details, hitratio)

Write the relational algebraic expression for the following tasks:

- i. Find all (queryno, uid) pairs for query with a hitratio value greater than 500.
- ii. Find all item names of items in Pokhara ordered with query_details as pokhara_details.
- iii. Find itemids of items ordered through manager 35 but not through manager 27.

- b) Write SQL statements for following:

- i. Create a table named Vehicle with veh_number as primary key and following attributes:
veh_type, veh_brand, veh_year, veh_mileage, veh_owner,

veh_photo, veh_price

- ii. Enter a full detailed information of a vehicle.
- iii. Increment vehicle's price by 10,000.
- iv. Remove all vehicle's records whose brand contains character 'o' in second position.
- v. Display the total price of all vehicles.
- vi. Create a view from above table.
- vii. Display details of vehicles ordering on descending manner in brand and by mileage when brand matches.
- viii. Change data type of year to datetime.

3. a) How does normalization help in organizing records in database? Justify with examples. 8
b) Write down the properties of decomposition. Compare & contrast assertion & triggers. 7
4. a) Differentiate between authorization and authentication. Explain about access control and view. 7
b) What is query optimization? List some strategies for optimization of queries and explain steps in for query processing with necessary diagram. 8
5. a) What is file organization? Explain how you organize files using B+ tree and hash index. 8
b) What do you mean by crash recovery? Differentiate between deferred database modification and immediate database modification. 7
6. a) Define transaction & schedule. Explain different states in a transaction. 7
b) Explain about distributed databases with its advantages and disadvantages. 8
7. Write short notes on: (Any two) 2×5
 - a) Sequential File Organization
 - b) Cascading in referential integrity
 - c) Data warehouse & Data mining

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2019
Programme: BE		Full Marks: 100
Course: Database Management System		Pass Marks: 45
		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the concept of DBMS and its applications tracing the evolution. 7
- b) Construct an ER diagram for keeping records for Library Management Systems. 8
2. a) Using the following schema represent the following queries using Relational algebra : 8
 - PROJECT (Project num, ProjectName, ProjectType, ProjectManager)
 - EMPLOYEE (Empnum, Empname)
 - ASSIGNED_TO (Projectnum, Empnum)
 - i) Find Employee details working on a project name starts with 'L'
 - ii) List all the employee details who are working under project manager "Rohan"
 - iii) List the employees who are still not assigned with any project.
 - iv) List the employees who are working in more than one project.
- b) Write the SQL statements for the following queries by reference of **Hotel_details** relation: 7

hotel_id	hotel_name	estb_year	hotel_star	hotel_worth
1	Hyatt	2047	Five	15M
2	Hotel Ktm	2043	Three	5M
3	Fulbari	2058	Five	20M
4	Yak and Yeti	2052	Four	11M
5	Hotel Chitwan	2055	Three	7M

- i. Create a database named hotel & table relation.
- ii. Create a view named Price which shows hotel name & its worth.
- iii. Modify the data so that Hotel Chitwan is now four star level.
- iv. Delete the records of all hotels having worth more than 9M.

3. a) What are store procedures? Explain equi Join, natural join, left and right outer join with examples. 8
- b) Differentiate between Functional Dependency and Multi Valued Dependency? Explain closure set of functional dependencies with example. 7
4. a) Define third normal form. Convert the following 2NF relation into 3NF(consider **Name** as primary key) 8

<u>Name</u>	Address	Phone	Salary	Post
Gill	KTM	456789	20000	Engineer
Van	BKT	654321	20000	Engineer
Robert	KTM	456789	20000	Engineer
Brown	BKT	654321	10000	Overseer
Albert	KTM	454545	10000	Officer

- b) What is security and integrity violations? Explain the need of access control, Authorization and Authentication. 7
5. a) What is query cost estimation? Explain cost based & heuristic based choice of evaluation plan for query optimization. 7
- b) Create a B+ tree of order 4 with following data: 8
(4, 9, 16, 25, 1, 20, 13, 15, 10, 11, 12) of order 4. Assume that, tree is initially empty and values are added in ascending order.
Also, show the formation of tree after the deletion of 16.
6. a) What is concurrency control? Describe ACID property of transaction. 8
- b) Define recovery. When the two transactions are said to be in deadlock state? How these deadlocks can be addressed. 7
7. Write short notes on: (**Any two**) 2×5
- a) Architecture of Distributed Database
- b) Role of Database administrator
- c) Dense and Sparse Index

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System

Semester: Fall

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why data independence is important in data modeling? Differentiate between physical and logical data independence. 7
b) Define and explain benefits of data model. Draw an E-R diagram for a Vehicle Management System including primary key, weak entity, composite attribute, derived attribute and multivalued attributes in your ER diagram. 8
2. a) Explain Relational Algebra. What are the relational algebra operations that can be performed? Give an example of all. 7
b) Write SQL statements for following: 8
 - i. Create a table named Automotor with chasis_number as primary key and following attributes: veh_brand, veh_name, veh_model, veh_year, veh_cost, veh_color, veh_weight
 - ii. Enter a full detailed information of an automotor.
 - iii. Change any Automotor's year to 2019.
 - iv. Remove all Automotor records whose model contains character 'i' in last position.
 - v. Display the total cost of all vehicles of the table Automotor.
 - vi. Create a view from above table having vehicles only red color.
 - vii. Display details of Automotor ordering on descending manner by brand name and by ascending on model when brand matches.
 - viii. Change data type of color so that it only takes one character.
3. a) Differentiate between join and sub query. Explain different SQL joins with examples. 8

- b) What is functional dependency? Discuss its types. Explain the role of Functional dependency in the process of normalization. 7
4. a) What is multi-valued dependency? Illustrate the advantage of 4NF with suitable example. 8
- b) Describe the GRANT functions and explain how it relates to security. What types of privileges may be granted? How rights could be revoked? 7
5. a) Define query optimization. What are the basic steps of query processing? Explain. 7
- b) In terms of file organization, define *Indexing*, *Elevator Algorithm*, *Log disk*. How does a mechanical hard disk work? 8
6. a) What is a transaction? What is a serializable schedule? Describe the dead lock handling mechanism. 7
- b) Explain different types of crash recovery algorithm with suitable examples. 8
7. Write short notes on any two: 2×5
- a) Two phase locking
- b) Data Godown v/s Data Warehouse
- c) Schema and instances